

AHA/ACC Synthetic Clinical Guideline: HFrEF Pharmacotherapy

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Section 1: Initial Clinical Assessment and Quadruple Therapy

For patients diagnosed with symptomatic Stage C Heart Failure with Reduced Ejection Fraction (HFrEF, LVEF $\leq$ 40%), guideline-directed medical therapy (GDMT) comprises quadruple pharmacotherapy initiated sequentially or in rapid succession. The foundational pillars include an ARNI/ACEi, evidence-based beta blocker (Carvedilol, Metoprolol Succinate, or Bisoprolol), a Mineralocorticoid Receptor Antagonist (MRA), and an SGLT2 inhibitor.

Section 2: Pharmacotherapy Dosing and Titration Table

Drug Class	First-Line Agent	Starting Dose	Target Maintenance Dose
ARNI	Sacubitril/Valsartan	49/51 mg PO BID	97/103 mg PO BID
Beta-Blocker	Carvedilol	3.125 mg PO BID	25 mg PO BID (50 mg if >85kg)
MRA	Spironolactone	12.5 - 25 mg PO Daily	25 - 50 mg PO Daily
SGLT2i	Empagliflozin	10 mg PO Daily	10 mg PO Daily

Section 3: Renal Impairment and Hyperkalemia Monitoring

Monitor serum creatinine, estimated GFR, and serum potassium at 1 to 2 weeks post-initiation or dose escalation. A transient rise in serum creatinine of up to 30% from baseline is acceptable and reflects intraglomerular hemodynamic changes. Withhold or down-titrate MRA if serum potassium exceeds 5.5 mEq/L or eGFR drops below 20 mL/min/1.73m².